

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICALPROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	AIDS-2024	Date:	

Code of Conduct

I J. Pranitha(192472073) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: PRANITHA

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2

Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

ANSWER:

Input Word	Root	Suffix	Type	Normalized Form
connected	connect	-ed	Inflectional (Past Tense)	connect

connecting	connect	-ing	Inflectional (Present Participle)	connect
connection	connect	-ion	Derivational (Noun Formation)	connect

Parsed Structure

- connected → connect + ed
- connecting → connect + ing
- connection → connect + ion

Normalized Output: connect

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

ANSWER:

Input Word	Prefix	Root	Suffix	Type	Morphological Breakdown
unhappy	un-	happy	—	Derivational	un + happy
happiness	—	happy	-ness	Derivational	happy + ness

happily	—	happy	-ly	Derivational	happy + ly
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Semantic Consistency

- Base meaning: happy
- unhappy → opposite meaning
- happiness → state of being happy
- happily → manner of being happy

Common Root: happy

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for

a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

ANSWER:

Input Word	Rule Applied	Stem	Type
played	Remove -ed	play	Inflectional
player	Remove -er	play	Derivational
playing	Remove -ing	play	Inflectional

Root Extraction

- played → play + ed
- player → play + er
- playing → play + ing

Final Stem: play

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset. Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

ANSWER:

Input Word	State Transition	Morphological Breakdown	Pattern	Normalized Form
writes	write → write + s	Root + Present Suffix	Regular Inflection	write
writing	write → write + ing	Root + Progressive Suffix	Regular Inflection	write
written	write → written	Root + Irregular PastParticiple	Irregular Inflection	write

State Transitions

- **writes**: Start → Root(write) → Add -s → Accept
- **writing**: Start → Root(write) → Add -ing → Accept
- **written**: Start → Root(write) → Irregular transition → **written** → Accept

Common Root: write

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.

Handle variations in suffix patterns while preserving semantic consistency.

Ensure uniform root extraction across all derived forms.

Determine the stemming steps, intermediate forms, and final stem for each input word.

ANSWER:

Input Word	Stemming Step	Intermediate Form	Final Stem
relational	Remove -ational → -ate	relate	relat

relation	Remove -ion	relat	relat
relate	Remove final -e	relat	relat

Final Output:

Input	Final Stem
relational	relat
relation	relat
relate	relat

Common Porter Stem: relat
Rubrics for Calculation

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem				
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects method				
Solution Process	25%	Logical, systematic steps with correct approach throughout	Most steps with minor mistakes				
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors				

Q. No. /Criteria Level	Problem Understanding	Method Selection	Solution Process
1			
2			

3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____